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Cited Text Span Extraction for Scientific Question Answering

Bachelor's Thesis Expose

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1. Introduction

Large Language Models (LLMs) have fundamentally transformed how we look up information and obtain answers to questions, becoming an increasingly important tool, not only in everyday life, but also across a range of professional fields. With the usage of such systems, the need to verify the factual correctness and easily do further research on the underlying sources arises, especially when answers lack important details or provide a depth of explanation that is misaligned with the user's existing knowledge. This is especially important in high-stakes fields such as medicine, law or scientific research, where seemingly plausible but unbacked or false claims could lead users to draw false conclusions.

One increasingly popular method is the approach of Retrieval Augmented Generation (RAG), replacing zero-shot LLM solutions with a two-step process. Firstly, relevant sources in the form of documents or other multi-media data are retrieved and then fed into the generating model for it to use as ground truth for answer generation and later to enable citation extraction. Retrieving correct and precise text-spans alongside the generated answer is not a trivial task, and with domain-specific requirements widely varying, such systems are part of active research [1] [2]. One system applying such a RAG approach is SQuAI [3], aimed at answering complex open-domain scientific questions on its knowledgebase containing 2.3 million full-text scientific papers. For further performance improvements, it introduces a multi-agent pipeline, enhancing document retrieval capabilities as well as providing document-level backlinks for parts of its generated answer. This system, while providing some backlinking, does still fall short of providing more granular, sentence-level cited text spans, which have proven to be much more useful to users, as they are quicker and easier to verify, enhancing trustworthiness and usability of the system.

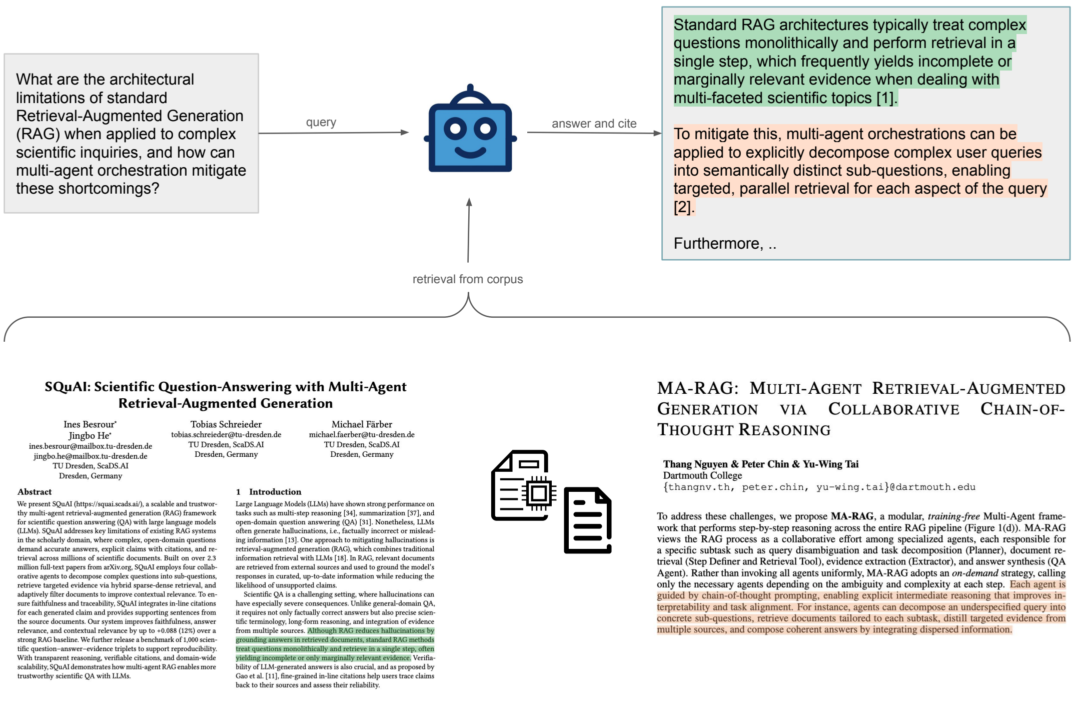

Figure 1.1.: Sentence level cited text span attribution

Therefore, this thesis will focus on benchmarking different cited text span extraction methods to enhance source attribution, leading to the following core contributions:

1. Curate a dataset of high-quality, multi-hop scientific questions building upon a large-scale retrieval corpus that require complex reasoning capabilities in order to be answered and closely resemble real-world scientific inquiry needs.

2. Introduce a hierarchy of different cited text span extraction methods ranging from simple lexical and semantic matching up to complex reasoning capabilities, as well as strategic combinations of such, to allow for tradeoffs in accuracy and efficiency.

3. Apply an automated framework to benchmark the retrieval and answering capabilities of non-parametric, retrieval- and evidence-based text generation systems on the proposed dataset as well as evaluate the extraction methods in terms of efficiency, correctness and quality of source attribution using NLI models and LLM-as-a-judge approaches.

2. Related Work

This chapter provides a comprehensive review of prior work across three key dimensions relevant for the development of a scientific QA system capable of providing cited text spans as evidence for its generated answer. First, it examines existing scientific QA datasets with respect to their suitability for SQuAI. It then compares different established cited text span extraction paradigms and their tradeoffs in terms of precision, efficiency and computational requirements before discussing automatic approaches for evaluating text span attribution quality in long-form, multi-hop settings.

2.1. Datasets

Utilizing SQuAI as a system basis, this work conducts a systematic evaluation of cited text span extraction methods for fine-grained source attribution in scientific QA. Therefore, a highly specialized dataset of open-domain, long form scientific questions is needed. Due to the scarcity and high cost of curating such datasets, only a limited number of suitable options are available. With SQuAI being designed as a system to answer complex scientific inquiries based on the unarXive 2022 Dataset [4], further requirements are introduced, namely:

1. Domain Alignment: Questions must fall under the knowledge scope of the underlying knowledgebase, in order for the system to be able to provide cited text spans as proof alongside its answer. In terms of the SQuAI System, this boils down to questions targeting physics, mathematics and computer science that are also part of the papers in the unarXive 2022 Dataset.

2. High Complexity: Questions must go beyond simple factoid questions and require complex reasoning capabilities to answer, as questions posed by scientists usually require synthesizing, interpreting and evaluating data, requiring high cognitive abilities to interact with. In order to be able to test if the SQuAI system can be used under such circumstances, the questions in the dataset must also have a similar degree of complexity.

3. Multi-Hop: Questions must require the information in multiple documents to be answered, in order to test the ability of the system to synthesize information from multiple sources, as is required in naturally occurring scientific research.

4. Decontextualization: Questions must be answerable in an open-domain setting and cannot contain any kind of reference to specific papers and their contents that lose meaning or become ambiguous outside of the context of their paper. As a result, questions must contain enough context such that they are standalone answerable.

5. Text format: Questions can only target information in a paper that is present in the form of text, as current versions of the SQuAI system do solely rely on textual information, not taking into account images, graphs or similar.

6. Ground Truth: In order to evaluate the correctness of the model's answer, a dataset must provide a ground answer alongside each question.

However, comparing some of the most prominent datasets shows that none fits all discussed criteria, finally resulting in the need for a newly curated dataset.

Dataset	**Domain Alignment**	**High Complexity**	**Multi-Hop**	**Decontextual-ization**	**Text format**	**Ground Truth**	**Sufficient Dataset size (>500)**
Qasper [5]	partial	shallow	no	no	yes	yes	yes
ALCE [6]	no	shallow	yes	yes	yes	yes	yes
SPIQA [7]	yes	complex	no	partially	no	yes	yes
QASA [8]	yes	complex	no	no	yes	yes	yes
ExpertQA [9]	yes	very complex	yes	yes	yes	yes	no [1]
SciDQA [10]	yes	very complex	no	no	no	yes	yes
PeerQA [11]	yes	very complex	no	no	yes	yes.	yes

Table 2.1.: Comparison of Scientific QA Datasets against SQuAI Requirements
Abbreviations: CS = Computer Science; ML = Machine Learning; NLP = Natural Language Processing.

2.2. Cited Text Span Extraction

Context extraction methods can be most broadly divided into pre- and post-hoc extraction methods, depending on the time the context attribution happens. While pre-hoc systems such as QASPER [5] and QASA [8] extract citable passages before and during runtime of the question-answering system, post-hoc systems search and attach supporting evidence afterwards. Post-hoc systems have been shown to outperform pre-hoc systems, achieving higher citation coverage and answer correctness as well as lowering model hallucination [12]. While a number of systems rely on zero-shot LLM solutions for context extraction [13] [14] [15], work on the DECOMPTUNE [16] system has shown that decomposing facts and decontextualizing answer sentences before attributing source text spans with LLM prompting can yield significant improvements, especially when applied to answers for long-form, multi-hop questions. In order to eliminate the probability of hallucinated model output, models can be finetuned [17] or the answer space restricted to the start and end token of the cited text span, forcing strict citation [18].

So-called white-box methods rely on access to the model's internal weights, rendering them incompatible with proprietary LLMs. The LAQuer [19] system, as well as earlier token-embedding approaches it is built upon [20], analyzes internal hidden states to extract the most likely text span that a user-highlighted part of the LLM's answer is based upon. To pinpoint a certain text span, the system calculates the maximum cosine similarity between individual tokens in the source text and the averaged hidden representation of the highlighted span, for which attribution is sought. The system then extracts the precise quote from the surrounding window of text. However, this method has been shown by the authors to perform significantly worse than LLM Prompting, with drastically lower accuracy and missing suitable citations, especially in larger search spaces, e.g.

[1] question-answer pairs are broadly distributed across diverse fields of study, resulting in insufficient topical overlap with unarXive 2022

longer text. Similarly, AttnAttribute [21] identifies "Attention Heads", tracking the model's internal attention values during generation time. Just like LAQuer, it then extracts text spans from the passage surrounding the anchor with the highest model attention. Benchmarks made as part of the development of DECOMPTUNE [16] concluded that both of those methods performed significantly worse than LLM Prompting.

Recent advancements have explored generative approaches. Rather than retrieving and reranking text spans, such systems generate them. In order to prevent hallucinations, the output vocabulary of the systems is typically restricted to the underlying retrieval corpus by applying constrained decoding mechanisms, such as prefix trees or FM-indices [22]. GERE [23], as one such system using a prefix tree, applies this concept of restricted output in the document retrieval step, generating document titles for selecting appropriate source texts, as well as sentence identifiers mapped to individual sentences. This system, while removing the need for a massive document index as well as showing significant improvements in precise and correct text span extraction compared to baselines such as Dense Passage Retrieval (DPR) and BM25, requires the prefix tree to be precomputed and stored in memory, making it highly inflexible for changes in the underlying corpus. Similarly, MRR-FV [24] further improves this approach with multi-hop reasoning capabilities. Instead of generating sentence identifiers, it generates sentences directly, leveraging the natural language understanding capabilities of LLMs, leading to improvements in precision and quality of attributed spans, but also further bloating the prefix tree, making the system even harder to scale.

Despite the diversity of proposed paradigms, current work mostly evaluates them in isolation, restricting comparison to baselines or direct predecessors on heterogeneous tasks, making broad comparisons for practical applicability unfeasible. Consequently, this work introduces a hierarchy of extraction methods, increasing in both reasoning capabilities and computational requirements, benchmarking both accuracy and efficiency on a unified framework, allowing for systematic trade-off analysis between paradigms in the context of open-domain multi-hop scientific QA.

2.3. System Evaluation

While human annotation has always been the gold standard for assessing natural language generation (NLG) tasks, relying solely on it is a bottleneck for developing models of increasing size and complexity. While providing feedback with extensive reasoning and natural language understanding, human evaluation suffers from being costly, time-intensive and, depending on the specific task, highly subjective. Thus, lowering or even replacing the need for human evaluation completely is an important problem.

Early metrics heavily rely on measuring the lexical overlap between text spans, for instance of a human-written ground truth and an LLM-generated answer for the same questions. The most prominent examples include BLEU [25] and ROUGE [26], both being n-gram based metrics. While the former is precision-oriented, measuring how many of its n-grams also appear in the human reference span, the latter applies a recall-oriented approach, measuring how many n-grams of the human reference are also present in the LLM-generated text span. Being computationally inexpensive and language independent, these metrics have established themselves as rigid industry standards and baselines for other methods. Their alignment with human judgement, however, has been shown to be low, suffering from the lack of semantic understanding, leading to penalizing valid paraphrasing as well as missing contradictory or hallucinated content.

Partly overcoming those limitations, embedding-based metrics like BERTScore [27] and MoverScore [28] emerged, measuring semantic instead of lexical similarity. BERTScore does one-to-one greedy matching on vector embeddings, providing precision, recall and F1 score metrics, just like BLEU and ROUGE, but on an embedding level. MoverScore improves upon this idea, utilizing Earth Mover's Distance (EMD) to allow for introducing one-to-many relationships, allowing for matching of single words with phrases, so-called soft matching, as well as introducing Inverse Document Frequency (IDF) for enhanced lexical matching, to better reflect the importance of specific keywords to be included. Though significantly more expensive to compute, those metrics show significant improvements in semantic robustness as well as intrinsic context awareness, due to their vector embeddings. However, both of those metrics still require a reference answer to compare to. BartScore [29] eliminates this requirement, measuring the conditional probability that one text span is generated from another using an encoder-decoder model, arguing that the probability is higher if the claims in the answer are also deeply rooted in the source text.

Besides similarity-based metrics, other research suggests approaching the task from a logical entailment perspective. The AIS [30] framework proposed a unified conceptual definition for such, arguing that a statement is considered Attributable to Identified Sources (AIS) iff completely corroborated by a source. In order to verify this, the authors imposed additional constraints being 1) that the claim made in an answer sentence must be standalone interpretable, thus decontextualized and 2) the information in the text span must support all the information present in the answer span it is attributed to. Previously discussed similarity-based metrics like BLEU, ROUGE or token-level F1, as well as solely relying on an out-of-the-box Natural Language Inference (NLI) model, have been shown to have low correlation with human judgement for both factual consistency and logical entailment assessment [31] [32]. To address these limitations, systems relying on NLI models like FactCC [32] argued that evaluating logical entailment on sentence-to-sentence level is insufficient, since information in sources might be spread across multiple passages. As a result, these systems increased their evaluation scope to the whole source text, leading to better judgement capabilities, but also increase in model input. The authors of FactCC observed that a majority of misclassifications were commonsense mistakes; the TRUE benchmark [31] observed improvements with increasing model size but also degradation with longer input, which is further supported by research on the SummaC [33] framework. To shrink input size, SummaC increased evaluation granularity by dividing both the source text and the model answer into smaller units, checking each combination of them for entailment and combining the obtained scores.

The Q^2 system [34] proposed using a Question-Generation - Question-Answering (QG-QA) method, generating questions to which the LLM-generated answer is the solution. Those questions are then answered with the original source document and lastly the so obtained answers are then compared with the original LLM answer using an NLI model. The authors presented significant improvements compared to baseline methods, further enhanced by the TRUE benchmark by applying an ensemble approach utilizing ANLI (standard NLI), SummaC (increased granularity NLI) as well as the Q^2 (QG-QA) method. The ensemble was shown to beat every standalone evaluation metric previously discussed on all datasets benched in TRUE, emphasizing the need for combined, complementary evaluation metrics.

Recently, the evaluation paradigm has shifted to LLM-as-a-judge approaches. While early versions like GPTScore [35] tried evaluating based on model internals, using the conditional token probability, thus the confidence of a model in a given answer, they have quickly been shown to be much less effective than prompting the LLM directly to grade [36]. The performance of LLMs in evaluation tasks can additionally be enhanced by applying a Chain-of-thought approach, prompting the model to break down complex tasks into simpler ones and adding intermediate reasoning steps. Closely resembling human cognitive processes, this method yields significant improvements, especially with larger models and problems of increasing complexity [37], reaching agreement rates comparable to human judges for open-ended multi-turn dialogues [38], thus eliminating human annotation constraints.

When applied to long-form text containing multiple pieces of information, binary judgements become insufficient as they cannot reflect partial attribution. FactScore [39] addresses this by breaking down answers into atomic facts using an LLM, obtaining a percentage measure of supportedness. AttrScore [40] introduces additional evaluation criteria, such as measures for attribution, extrapolation and contradiction, in order to further characterize the type of attribution error. Similarly, CiteEval [41] introduces penalties for attributions containing unnecessary noise and redundant citations while advocating for evaluation against full source documents, in order to not lose information in the surrounding context. The RAGAS [42] framework operationalized these findings, providing a suite of automated metrics like faithfulness, answer relevance and context relevance, without requiring human annotation. However, the importance of penalizing undesired behaviors is further emphasized by the MT-Benchmark [38], showing that LLMs have a multitude of biases, such as favoring longer and LLM-generated text spans or switching preferences between answers when presented in a different order. Furthermore, LLMs perform remarkably poorly with fine-grained information, with a majority of mistakes being caused by overlooking numbers and entities or hallucinating logical connections [43]. Using increasingly large models additionally introduces high computational cost requirements. To address this, systems like ARES [44] rely on LLM-generated, domain-specific, synthetic training data to finetune smaller models. Using Prediction-Powered-Inference (PPI), ARES leverages a small human-annotated dataset to calculate the LLM's annotation error on the synthetic data, the "rectifier", to provide statistical confidence bounds and enhance its evaluation, beating existing zero-shot solutions like RAGAS.

Automatic evaluation of cited text span attribution remains a challenging task. Despite the wide range of approaches, no single metric being capable of reliably and comprehensively assessing attribution performance as a whole has emerged. Increasingly complex tasks, such as multi-hop questions and challenging application domains like scientific QA, introduce further evaluation dimensions. Consequently, combining multiple evaluation metrics with complementary strengths becomes necessary to measure task-specific aspects within a unified framework.

3. Methodology and Approach

3.1. Dataset

To curate a dataset of high-quality, expert-level scientific question-answer pairs, a fully automated pipeline is applied. It consists of three consecutive steps: document selection, question answer generation and quality assurance, in order to ensure that generated questions and answers are scientifically rigorous and strictly dependent on multi-document information synthesis to be answered, as well as reliably providing associated ground truth documents alongside.

3.1.1. Document selection

To match the underlying document corpus, selectable papers are restricted to those of the unarXive 2022 dataset, yet for other application this could be expanded at will. A randomly selected anchor paper as well as its cited references serve as the foundation for generating a single question-answer pair. For each paper, a document-level SPECTER [45] vector embedding is retrieved via the Semantic Scholar academic graph API [46]. Leveraging citation graphs, these vectors provide dense representations of a paper's title and abstract, which are subsequently used to calculate the cosine similarity and thus the semantic similarity of each paper with the anchor document. To ensure topical alignment while excluding highly redundant work, a subset of candidate papers with a cosine similarity between 0.7 and 0.9 is selected. Four of these papers are then chosen at random to be used alongside the anchor paper for generation. Restricting the model input to five papers in total provides a multitude of topical overlaps for the model to choose from without resulting in exceedingly long prompts, mitigating the risk of "Lost in the Middle" effects. If a chosen anchor paper fails to provide at least four appropriate references, it is considered insufficient to be used in the dataset.

3.1.2. Question-Answer generation

While modern LLMs offer increasingly large context windows, usually even above 100,000 tokens, empirical studies have shown severe performance degredation in tasks like complex reasoning, data retrieval and logic problems, accompanied by drastically increased hallucination [47]. In the generation step, the pipeline applies a moonshotai/Kimi-K2.6 [48] model, supporting a context window of approximately 256.000 tokens. In practice however, the "effective context window", in which models only suffer from minimal performance degredations with increasingly long inputs, has been show to drastically lower by emipirical studies [49] and practitioner reports, with some studies estimating it as low as 40% of the maximum supported context window [50]. To ensure high reasoning capabilities, the pipeline applies an adaptive character limit for each paper text, enforcing a total upper bound of 125.000 characters for all papers combined which conservatively approximated translates to roughly 32.000 tokens[2] , well below Kimi-K2-6's effective context. Papers are assigned input

[2] in standard heuristics for scientific english one token translates to roughly three to four characters

space according to the following formula, allowing for for full and efficient utilization of the context window, even if papers are highly variable in length:

$$C_{adaptive} = \left\lfloor C_{base} + \left(\frac{l_{paper}}{\sum_{p \in P_{above}} len(p)} \right) \times \sum_{s \in P_{below}} (C_{base} - len(s)) \right\rfloor \tag{3.1}$$

Paper texts longer than their adaptive character limit are then head and tail truncated, retaining the first and last $\lfloor \frac{C_{adaptive}}{2} \rfloor$ characters, which are the most information dense for scientific papers.

Subsequently the LLM is prompted using a two-step strategy. Firstly the model is tasked with selecting the best pair of papers (A,B), where paper A provides an intermediate entity, method, dataset or similar and paper B evaluating, extending, contradicting or relying on A's intermediate element. Following that, the LLM is tasked with generating a single Question-Answer pair based on its selected pair, whose answer can only be derived by following a sequential reasoning process across both papers, applying paper A's intermediate element to paper B. Additionally, the model is instructed to answer the question as precisely as possible, providing a precision-biased subset of the underlying papers used during generation alongside.

3.1.3. Quality assurance

In order to ensure the dataset's integrity two separate evaluation steps are applied, both using an openai/gpt-oss-120b [51] model. The first judge evaluates each Question-Answer pair along six distinct dimension, providing judgement on how good each criteria is satisfied on a three-point ordinal scale.

1. **Multi-Hop validity** assesses whether answering the question requires integrating information from both papers. Thus, this criteria is only satisfied, if neither paper provides sufficient information for deriving the complete answer. A question only satisfied this criteria if neither paper alone provides sufficient evidence to derive the complete answer.

2. **Dependency Strength** evaluates whether the steps in the reasoning chain are sequentially dependent, in order for the conclusion to be drawn. Thus, the second reasoning step cannot be solved without first applying or interpreting the first one.

3. **Non-decomposability** verifies, that questions cannot be broken down into independent single-hop sub-questions, allowing for reasoning shortcuts or simply combining sub-question answers. Instead, reasoning steps must be connected in such a way, that separately solving them would not be sufficient for deriving an answer.

4. **Evidence distribution** examines whether both papers contribute distinct and necessary evidence, ensuring that required evidence is spread across both papers and questions cannot be answered by using a single dominant source, with redundant information.

5. **Answerability** assesses whether the answer is fully supported and obtainable by only relying on information present in the papers, without requiring external knowledge or speculative inference, not explicitly supported by either paper. Is the answer fully supported by the provided evidence?

6. **Decontextualization** validates, that the question is self-contained and unambiguous in an open-domain setting. Thus, questions cannot contain context specific references and must introduce instead relevant concepts, methods or entities by themselves, in order to satisfy this criteria.

An analysis of disconnected reasoning by Trivedi et al. [52] demonstrated, that many alleged multi-hop question-answering datasets do in fact not require true multi-hop reasoning, but can instead be solved through reasoning shortcuts, partial evidence or decomposition into independent sub-questions. Using a methodology inspired by the DIRE [52] framework, the second judge verifies, that questions cannot be answered under incomplete evidence, namely a subset $P' \subset P$ of ground truth papers P, but instead require all ground truth documents for a complete reasoning chain and answer. Thus the test outcome satisfies:

$$(\forall P' \subset P : \neg \mathrm{Pass}(P')) \wedge \mathrm{Pass}(P)$$

The final dataset is then obtained by filtering question-answer pairs to only those, successfully passing both judges. In order to verify the pipeline and the judges, human verification will be applied on a subset.

3.2. Cited text span extraction

Constrained by practical limitations of the granularity of attribution by the SQuAI system, providing exactly one source document per answer sentence, the source span extraction task in this work is formulated as a post-hoc attribution step of providing one source span per answer sentence. The retrieval methods themselves however, are applicable to other systems with alternative granularity of attribution. This constitutes a limitation for this specific system, especially in cases where one attribution scope (e. g. a sentence) contains multiple claims requiring evidence, but also avoids additional ambiguity introduced by finer attribution granularity. For the purpose of comparing different extraction methods under identical conditions and constraints, however, this granularity is sufficient in order to measure whether extraction variants are capable of identifying relevant evidence, with results being directly comparable.

As previously discussed, cited text span attribution can be approached through a number of different paradigms, with methods being introduced and applied under vastly different circumstances. Following, a hierarchy of increasingly complex post-hoc extraction methods that will be applied on a unified dataset is introduced. All methods except the native SQuAI span extraction and the generative LLM method receive tuples, composed of an answer sentence and a set of eligible source spans $\mathcal{W}(P)$ from the paper P, annotated as factual basis for that sentence during question answering. The set of source spans is composed of floating windows of contiguous sentences in the source document, up to a length of five, and can be denoted as:

$$\mathcal{W}(P) = \bigcup_{k=1}^{5} \{s_i \oplus s_{i+1} \oplus \cdots \oplus s_{i+k-1} \mid 1 \leq i \leq n-k+1\},$$

where k specifies the window size and $\oplus$ denotes string concatenation with whitespace.

Extraction Method	Floating windows	Retrieval Category	Candidate Selection (Top 10)	Final Selection (Top 1)
1	×	Native	-	SQuAI
2	✓	Dense	-	Bi-encoder
3	✓	Sparse	-	BM25
4	✓	Hybrid	Bi-encoder	BM25
5	✓	Hybrid	BM25	Bi-encoder
6	✓	Hybrid	-	BM25 & Bi-encoder [3]
7	✓	Cross-encoder	Bi-encoder	Cross-encoder
8	✓	Cross-encoder	BM25	Cross-encoder
9	✓	Cross-encoder	BM25 & Bi-encoder [3]	Cross-encoder
10	×	Generative	-	LLM

Table 3.1.: Cited text span extraction methods overview

3.2.1. Native SQuAI extraction baseline

The first method relies on SQuAI's native text span extraction mechanism, utilizing simple lexical matching. Since it is only capable of providing one span per paper, even if it is cited repeatedly in different contexts, it is limited by design. Nevertheless, representing the attribution behaviour of the SQuAI system, it serves as a native baseline for further extraction pipelines, to compare against.

3.2.2. Dense retrieval

Applying a bi-encoder, this method selects the best fitting source span out of all floating windows based on the similarity of the span's dense vector representation with that of the answer sentence it is attributed to. It measures the suitability and sufficiency of relying exclusively on semantic similarity for source span attribution. As a dense retrieval baseline, it is capable of identifying appropriate source spans with different wording, but prone is to overlooking precise terminology.

3.2.3. Sparse retrieval

The third method utilizes BM25, reranking source spans based on their lexical similarity with the corresponding answer sentence. The highest scoring span is then selected as the final attributed span. Representing the sparse retrieval baseline BM25 is especially effective with precise terminology, however less robust to paraphrasing. Its purpose is to test wether focussing on specific terms, entities or important keywords alone proves effective for cited text span extraction.

[3] using Reciprocal Rank Fusion

3.2.4. Hybrid retrieval

Hybrid methods aim to combine sparse and dense retrieval, leveraging their distinct characteristics. Thus, these variants evaluate whether the retrieval performance can be enhanced by combining both approaches into a single method. The fourth method applies a two-stage retrieval method, with the bi-encoder selecting the top ten candidate windows and BM25 selecting the final span, with method five reversing their order. While the former prioritizes semantic recall during candidate selection and lexical precision during final span selection, the latter reverses this order. Comparing both allows for analysis, if either of those paradigms is better suited for candidate or final span selection. The sixth method combines both lexical and semantic matching into a single score using Reciprocal Rank Fusion (RRF). Instead of sequential application, this variant evaluates hybrid retrieval performance without an imposed order between the components themselves. Thus, this combination allows for assessment of whether the paradigms capture complementary aspects of the attribution task, resulting in performance enhancements when applied at the same time.

3.2.5. Cross-Encoder reranking

Overcoming the limitation of BM25 and bi-encoder retrieval processing the answer sentence and source span in isolation, the following methods introduce cross-encoders, allowing for joint processing of both. This promises improvements, especially for context windows where fine-grained semantic differences such as negations or conditions as well as important entities or numerical values determine whether the answer sentence is merely related, contradicted or truly supported. These variants therefore evaluate whether joint answer-span comparison translates into better evidence selection in the form of cited text spans. However, as cross-encoders are far more computationally expensive than previous methods and thus not feasible to run on the whole set of floating window candidate spans, these methods instead use different candidate selectors, with the cross-encoder selecting the best fitting final span for the answer sentence. Methods seven and eight apply semantic and lexical matching respectively, while method nine utilizes both joined together as a candidate selector. Comparing these variations, allows assessing which candidate selection strategy provides the best basis for a cross-encoder and therefore improves overall attribution quality.

3.2.6. Generative LLM

The last method uses a generative LLM to extract the best supporting span from the source document. Unlike previous methods, it does not rerank candidate spans, but instead prompts the model to identify the best suited text span directly. Representing a generative approach, it is used to test whether an LLM's reasoning capabilities can improve evidence selection beyond retrieval and reranking, serving as a high-cost reference method. The complete prompt instructions are provided in Listing

3.3. System evaluation

The performance of the text span extraction methods can only be evaluated conditionally on the outputs of preceding pipeline steps, since post-hoc attribution is heavily dependent on their results. The evaluation is therefore split into assessing the suitability of the upstream inputs for text span extraction and evaluating the quality of the extracted spans themselves. This separation prevents text-span evaluation results from being confounded by errors introduced before span extraction.

3.3.1. Upstream system evaluation

Upstream inputs, leading to bad span attributions, can be flawed for a number of reasons, including:

1. Retrieval failure: The system retrieves or filters the wrong or insufficient subset of papers.
2. Generation failure: The system generates a factually wrong, overgeneralized, hallucinated or insufficiently supported answer.
3. Document attribution failure: The system fails or incorrectly attributes source documents to answer sentences.

These must be addressed separately to evaluating the attributed spans.

Retrieval failure can be analyzed by comparing the retrieved documents with those used during question generation, making it possible to not only evaluate the retrieval capabilities of the SQuAI system itself using precision, recall and F1-scores, but also the conditional performance of attribution methods given only unrelated, partial or complete evidence sources.

Generation failure is examined by comparing the system-generated answer with the ground-truth answer provided in the dataset. This comparison evaluates whether the system's answer covers at least the factual content of the reference, including all relevant facts and without introducing contradictory information. Exact lexical overlap is not required, as the comparison is performed by an LLM-as-a-judge, whose reliability is validated through human annotation on a small subset.

Document attribution failure ...

3.3.2. Span evaluation

4. Timetable

The development of this thesis is based on extensive literature research conducted prior to this Exposé in order to properly outline the requirement and development scope of the dataset, the system architecture as well as the thesis itself. The initial setup on the High-Performance Computing Cluster has been completed. Currently, the literature search is nearing completion and the curation of the dataset is in progress, resulting in the following timeline for thesis completion:

Stage	**Timeframe**
Literature Search	20/04/2026 – 27/04/2026
System design	27/04/2026 – 04/05/2026
Implementation	04/05/2026 – 18/05/2026
Evaluation	18/05/2026 – 01/06/2026
Conclusion	01/06/2026 – 29/06/2026

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